

ALBERTUS, IL, USA

WMO: 722082

Lat: 42.246N

Lon: 89.582W

Elev: 262

StdP: 98.22

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04876

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-18.3	-25.7	0.4	-21.0	-22.8	0.5	-17.6	11.8	-3.7	11.0	-3.0	2.6	270	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.7	23.8	30.0	22.8	28.7	22.3	25.7	29.7	24.5	28.4	23.4	27.2	5.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.4	28.7	23.0	18.4	26.9	22.2	17.5	26.0	80.7	29.6	75.5	28.5	71.2	27.3	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.3	8.4	-25.4	33.8	3.9	2.2	-28.2	35.3	-30.5	36.6	-32.6	37.8	-35.4	39.4		
(5)			-25.7	27.2	3.6	1.6	-28.4	28.3	-30.5	29.2	-32.5	30.1	-35.2	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-6.1	-5.0	2.4	9.1	15.1	20.4	21.9	20.8	17.4	10.4	3.3	-3.3		
	DBStd	11.22	6.77	6.54	6.28	5.00	5.02	3.23	3.38	3.02	4.25	5.08	5.48	6.03		
	HDD10.0	1935	498	419	249	74	11	0	0	0	2	58	210	414		
	HDD18.3	3784	756	652	496	278	127	17	7	11	66	252	452	672		
	CDD10.0	1547	0	1	13	48	168	311	369	334	225	69	8	1		
	CDD18.3	354	0	0	1	2	26	77	118	86	39	6	0	0		
	CDH23.3	3241	0	0	7	36	266	678	1120	723	354	57	0	0		
	CDH26.7	864	0	0	0	3	66	174	353	174	84	8	0	0		
(14) Wind	WSAvg	3.7	4.2	4.1	4.2	4.6	4.0	3.4	2.7	2.4	2.9	3.7	4.2	4.0		
(15) Precipitation	PrecAvg	936	40	39	58	92	106	131	97	106	92	66	65	44		
	PrecMax	1154	85	95	122	155	226	297	192	240	203	135	136	109		
	PrecMin	685	12	5	16	40	15	38	18	22	23	11	0	12		
	PrecStd	116	19	26	39	37	56	68	39	53	44	41	39	23		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	14.9	24.7	26.8	31.6	32.5	35.1	32.8	31.6	27.6	19.9	14.4		
		MCWB	8.8	10.7	17.0	16.9	20.9	23.7	26.9	23.8	23.3	20.1	14.2	12.9		
	2%	DB	6.6	8.8	17.9	23.9	28.1	30.3	32.4	30.1	29.0	25.0	16.6	9.4		
		MCWB	4.8	6.5	13.5	15.5	19.2	22.4	25.7	23.4	21.8	17.7	12.6	7.8		
	5%	DB	4.0	5.3	15.2	21.0	26.4	28.8	30.0	28.4	27.2	22.1	14.1	6.6		
		MCWB	2.5	3.2	11.1	13.1	18.7	21.6	23.6	22.7	20.2	16.6	11.0	5.2		
	10%	DB	2.5	3.0	11.8	18.0	24.1	27.2	28.6	27.3	25.3	19.1	11.8	4.1		
		MCWB	1.6	1.7	8.3	12.3	17.7	20.5	23.0	21.7	19.1	15.1	8.9	2.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.4	11.2	17.7	19.0	23.1	25.9	28.0	26.4	24.6	22.0	15.7	12.9		
		MCDB	10.2	13.5	23.5	24.3	29.0	30.5	33.0	31.1	29.0	26.3	18.1	14.6		
	2%	WB	5.1	6.5	14.7	16.4	21.5	24.2	26.4	24.8	23.3	19.9	13.3	7.8		
		MCDB	6.6	8.7	18.1	21.5	26.5	28.5	30.4	28.4	27.1	23.2	15.9	8.7		
	5%	WB	3.0	3.6	11.9	14.9	20.1	23.0	25.2	23.6	21.9	17.6	11.3	5.3		
		MCDB	3.9	4.9	14.8	19.7	24.7	27.1	29.0	27.1	25.4	20.3	13.6	6.4		
	10%	WB	1.4	1.9	8.5	13.1	18.6	21.9	23.8	22.6	20.6	15.4	8.9	3.1		
		MCDB	2.3	3.2	11.9	17.5	23.0	25.5	27.6	26.0	23.8	19.0	11.7	4.2		
(35) Mean Daily Temperature Range	MDBR		7.7	8.2	9.3	11.6	11.4	11.0	11.0	11.0	12.1	10.9	8.8	7.3		
	5% DB	MCDBR	8.7	10.2	13.6	16.1	13.7	12.5	11.6	11.8	13.3	14.3	11.7	9.2		
		MCWBR	7.6	8.0	9.4	9.4	7.2	6.8	6.1	6.6	7.0	8.3	8.6	8.0		
	5% WB	MCDBR	8.5	10.0	12.2	13.8	11.6	11.0	10.3	10.0	11.3	11.6	11.0	8.9		
		MCWBR	7.5	8.2	9.2	9.1	7.1	7.0	6.1	6.6	6.8	8.7	8.9	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.282	0.300	0.332	0.380	0.407	0.422	0.439	0.415	0.387	0.339	0.298	0.281		
	taud		2.397	2.365	2.392	2.272	2.254	2.254	2.200	2.266	2.324	2.458	2.507	2.434		
	Ebn at Noon		867	905	913	887	869	853	833	843	842	850	846	837		
	Edh at Noon		78	95	106	128	135	135	141	128	112	87	71	70		
(44) All-Sky Solar Radiation	RadAvg		1.73	2.72	3.73	4.64	5.43	6.13	6.09	5.38	4.45	2.91	1.90	1.39		
	RadStd		0.21	0.32	0.36	0.39	0.36	0.46	0.36	0.34	0.36	0.35	0.16	0.13		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days				
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page